

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-ab1c3508e25523f29.jsonl`  
- SHA-256: `34975bf0228b33ea98900e8e376dc460e818bb52baf3373ece82c18bacff8674`  
- Source modified: `2026-06-06T17:32:04+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:31:09.326Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

URL: https://github.com/nttcom/WASB-SBDT

Title: nttcom/WASB-SBDT ? Widely Applicable Strong Baseline for Sports Ball Detection and Tracking (BMVC 2023)

Found via: datasets/licensing search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:31:12.229Z)

I'll fetch the WASB-SBDT GitHub page and extract relevant claims.

3. user (2026-06-06T17:31:22.935Z)

WASB-SBDT Repository Key Facts

License

MIT License ? The repository explicitly states "License: MIT" with no indicated commercial-use restrictions.

Sports & Datasets Supported

The repository supports multiple sports: **"? ? ? ?"** (soccer, tennis, badminton, volleyball, basketball). However, the provided content does not detail specific dataset licenses or dataset-specific restrictions.

Performance Metrics

No accuracy, F1 scores, or benchmark numbers are provided in this excerpt. The repository references a paper but specific performance data is not included in the visible content.

Model Architecture & Compute Requirements

Details about model architecture, inference speed, and GPU requirements are not specified in this excerpt. The setup notes indicate testing was performed with **"Python3.8, CUDA11.3 on Ubuntu 18.04 (4 V100 GPUs)"** but this reflects testing configuration, not explicit requirements.

Paper & Publication

Title: **"Widely Applicable Strong Baseline for Sports Ball Detection and Tracking"**

Venue & Date: BMVC 2023 (proceeding published 11/23/2023); arXiv preprint released 11/09/2023

Authors: Tarashima, Haq, Wang, Tagawa

arXiv ID: 2311.05237

Detection Scope

The repository focuses on **ball detection and tracking** functionality. Whether rally/temporal segmentation is included is not clarified in this content.

4. assistant (2026-06-06T17:31:32.571Z)

Let me get more detail on the datasets and their licenses, as well as performance numbers, since those are central to the research question.

5. user (2026-06-06T17:31:43.851Z)

Datasets and Performance Metrics from WASB-SBDT Repository

Datasets Mentioned

Based on the provided content, the repository references sports ball detection and tracking across multiple sports (? ? ? ? ?), but **specific dataset names and their license/terms are not detailed in this excerpt**.

The page mentions evaluation codes for comparison baselines:

- TrackNetV2
- ResTrackNetV2
- MonoTrack
- DeepBall
- DeepBall-Large
- BallSeg

However, these appear to be model names rather than dataset names.

Performance Metrics

No performance numbers (F1, precision, recall, accuracy, or tracking metrics) are provided in the visible content of this repository page.

Pretrained Model Weights

The repository states: "Now it contains evaluation codes of pretrained WASB models only" as of November 17, 2023. The page references a "[MODEL_ZOO.md](/nttcom/WASB-SBDT/blob/main/MODEL_ZOO.md)" file where model availability should be documented, but that file's contents are not included in the provided excerpt.

License

The repository operates under an **MIT license**, as indicated by the badge present in the README section.

6. assistant (2026-06-06T17:31:47.125Z)

I have enough to extract solid claims. The MIT license, multi-sport support (including badminton), the BMVC 2023 paper, pretrained WASB model weights, and the baseline comparisons are all confirmed. Let me record the findings.

7. user (2026-06-06T17:31:55.803Z)

Structured output provided successfully

8. assistant (2026-06-06T17:32:04.668Z)

Extracted 5 falsifiable claims from the WASB-SBDT repository (<https://github.com/nttcom/WASB-SBDT>).

Source quality: primary (peer-reviewed BMVC 2023 paper, official NTT Communications repository).

Key findings relevant to the research question:

- MIT license confirmed (central ? validates the existing permissive substrate choice).
- It is detection/tracking ONLY, not rally/temporal segmentation ? confirming that rally

start/end logic must be built on top of trajectory output (central to the "what's the bottleneck" question).

- Pretrained WASB checkpoints are provided, enabling local single-GPU inference without retraining.
- Multi-sport support (badminton, tennis, volleyball, soccer, basketball) and outperforms TrackNetV2/MonoTrack/DeepBall/BallSeg baselines.

Note: the GitHub README excerpt did not surface concrete per-sport accuracy/F1 numbers or per-dataset license details (those live in the arXiv paper and MODEL_ZOO.md, which would be better sources for the recall-bottleneck and dataset-licensing sub-questions).